

- Description

This module manages an industrial steam pressure monitoring system, ensuring that:

1. **The steam pressure in a pipeline or boiler is continuously measured and maintained within a safe range.**
2. **If the pressure exceeds the maximum safety threshold, an overpressure warning is triggered.**
3. **If the pressure drops below the minimum threshold, a leakage or inefficiency alert is issued.**
4. **A log of all pressure readings, warnings, and corrective actions is maintained for system reliability.**

This system helps prevent dangerous overpressure conditions, detect potential leaks, and optimize steam efficiency.

- Specification

module IndustrialSteamPressureMonitoring

var currentPressure, maxPressure, minPressure: real

var log: sequence of string

process updatePressure(changeAmount: real)

functional scenario form:

currentPressure $\geq$ minPressure && log = log $\oplus$ ("Steam Pressure Updated: " + currentPressure + " bar"), Steam pressure updated successfully.

||

currentPressure > maxPressure && log = log $\oplus$ ("Overpressure Warning: " + currentPressure + " bar, Risk of System Failure."), Overpressure warning triggered, risk of failure.

||

currentPressure < minPressure && log = log $\oplus$ ("Leakage Alert: " + currentPressure + " bar, Efficiency Loss Risk."), Leakage alert triggered, potential efficiency loss.

end process

end module

- Java Code

```
package code;
import java.util.*;
class SteamSystem {
    private double maxPressure;
    private double minPressure;
    private double currentPressure;
    private List<String> log;
    public SteamSystem(double maxPressure, double minPressure, double initialPressure) {
        this.maxPressure = maxPressure;
```

```

        this.minPressure = minPressure;
        this.currentPressure = initialPressure;
        this.log = new ArrayList<>();
    }
    public void updatePressure(double changeAmount) {
        currentPressure += changeAmount;
        String message = String.format("Current Steam Pressure: %.2f bar", currentPressure);
        if (currentPressure > maxPressure) {
            message += " | WARNING: Overpressure Detected! Risk of System Failure.";
        } else if (currentPressure < minPressure) {
            message += " | ALERT: Possible Leak Detected! Efficiency Loss Risk.";
        }
        System.out.println(message);
        log.add(message);
    }
    public void printLog() {
        System.out.println("\nSteam Pressure Log:");
        for (String logEntry : log) {
            System.out.println(logEntry);
        }
    }
}

public class c38_IndustrialSteamSystem {
    public static void main(String[] args) {
        SteamSystem steamSystem = new SteamSystem(10.0, 3.0, 6.0);
        // testcase-VT:
        // steamSystem.updatePressure(3.5);
        // steamSystem.updatePressure(-4.0);
        // steamSystem.updatePressure(-3.0);
        // steamSystem.updatePressure(4.5);
        // steamSystem.updatePressure(4.0);

        //testcase-FT:
        // steamSystem.updatePressure(6.416398597203269E305);
        // steamSystem.updatePressure(-1.7976931348623157E308);
        // steamSystem.updatePressure(3.9229556063199447E298);
        // steamSystem.updatePressure(-6.838283965724573E307);
        // steamSystem.updatePressure(1.1092300594210293E296);
        // testcase-Z3:
        // steamSystem.updatePressure(3.5);
        // steamSystem.updatePressure(4.0);
    }
}

```

```
//      steamSystem.updatePressure(3.0);
//      steamSystem.updatePressure(4.5);
//      steamSystem.updatePressure(4.0);

// testcase-UVT:
steamSystem.updatePressure(3);
steamSystem.updatePressure(-40);
steamSystem.updatePressure(-30);
steamSystem.updatePressure(45);
steamSystem.updatePressure(4);

//      steamSystem.printLog();
}
}
```

- Coverage report

1. Test case number: 5
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	836	842
✓ src	0.7 %	6	836	842
✓ code	0.7 %	6	836	842
✓ c38_IndustrialSteamSystem.java	100.0 %	6	0	6
> c38_IndustrialSteamSystem		0	0	0
> SteamSystem	100.0 %	6	0	6

3. Vibration testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	106	20,036	20,142
✓ src	0.5 %	106	20,036	20,142
✓ code	0.5 %	106	20,036	20,142
✓ c38_IndustrialSteamSystem.java	97.2 %	106	3	109
> c38_IndustrialSteamSystem	89.3 %	25	3	28
> SteamSystem	100.0 %	81	0	81

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.6 %	5	837	842
✓ src	0.6 %	5	837	842
✓ code	0.6 %	5	837	842
✓ c38_IndustrialSteamSystem.java	83.3 %	5	1	6
> SteamSystem	83.3 %	5	1	6
> c38_IndustrialSteamSystem		0	0	0

5. Fuzz testing instruction coverage: 97.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	106	20,036	20,142
✓ src	0.5 %	106	20,036	20,142
✓ code	0.5 %	106	20,036	20,142
✓ c38_IndustrialSteamSystem.java	97.2 %	106	3	109
> c38_IndustrialSteamSystem	89.3 %	25	3	28
> SteamSystem	100.0 %	81	0	81

6. Z3 branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	837	840
▼ src	0.4 %	3	837	840
▼ code	0.4 %	3	837	840
▼ c38_IndustrialSteamSystem.java	50.0 %	3	3	6
> SteamSystem	50.0 %	3	3	6
> c38_IndustrialSteamSystem		0	0	0

7. Z3 instruction coverage: 76.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	82	22,300	22,382
▼ src	0.4 %	82	22,300	22,382
▼ code	0.4 %	82	22,300	22,382
▼ c38_IndustrialSteamSystem.java	76.6 %	82	25	107
> SteamSystem	72.8 %	59	22	81
> c38_IndustrialSteamSystem	88.5 %	23	3	26

8. Unimproved V-method branch coverage: 50%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	837	840
▼ src	0.4 %	3	837	840
▼ code	0.4 %	3	837	840
▼ c38_IndustrialSteamSystem.java	50.0 %	3	3	6
> SteamSystem	50.0 %	3	3	6
> c38_IndustrialSteamSystem		0	0	0

9. Unimproved V-method instruction coverage: 75.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	81	22,301	22,382
▼ src	0.4 %	81	22,301	22,382
▼ code	0.4 %	81	22,301	22,382
▼ c38_IndustrialSteamSystem.java	75.7 %	81	26	107
> SteamSystem	71.6 %	58	23	81
> c38_IndustrialSteamSystem	88.5 %	23	3	26